

NumericalOptimization 1

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PROBLEM I[2.1] Compute the gradient $\nabla f(x)$ and Hessian $\nabla^2 f(x)$ of the Rosenbrock function

$$f(x) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2.$$

Show that $x^* = (1, 1)^T$ is the only local minimizer of this function, and that the Hessian matrix at that point is positive definite.

SOLUTION. 令

$$r(x) = x_2 - x_1^2,$$

则 $f(x) = 100r(x)^2 + (1 - x_1)^2$ 。逐分量求导得

$$\nabla f(x) = \begin{pmatrix} -400x_1(x_2 - x_1^2) + 2(x_1 - 1) \\ 200(x_2 - x_1^2) \end{pmatrix},$$

并且

$$\nabla^2 f(x) = \begin{pmatrix} 1200x_1^2 - 400x_2 + 2 & -400x_1 \\ -400x_1 & 200 \end{pmatrix}.$$

由于 f 在 $\mathbb{R}^2$ 上连续可微, 任一局部极小点都必须满足一阶必要条件 $\nabla f(x) = 0$ 。由梯度的第二个分量为零可知

$$x_2 = x_1^2.$$

代回第一个分量, 得到 $2(x_1 - 1) = 0$, 因而 $x_1 = 1$, 进而 $x_2 = 1$ 。所以 $x^* = (1, 1)^T$ 是唯一的驻点, 也就是唯一可能的局部极小点。另一方面,

$$f(x) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2 \geq 0,$$

且等号成立当且仅当 $x_1 = 1$ 且 $x_2 = x_1^2 = 1$ 。故 x^* 实际上是唯一的严格全局极小点。

最后,

$$\nabla^2 f(x^*) = \begin{pmatrix} 802 & -400 \\ -400 & 200 \end{pmatrix}.$$

其顺序主子式满足

$$802 > 0, \quad \det \nabla^2 f(x^*) = 802 \cdot 200 - 400^2 = 400 > 0.$$

由 Sylvester 判别法, $\nabla^2 f(x^*)$ 正定。

□

PROBLEM II[2.5] Consider the function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x) = |0x|_{0_2}.$$

Show that the sequence of iterates $\{x_k\}$ defined by

$$x_k = \left(1 + \frac{1}{2^k}\right) \begin{pmatrix} \cos k \\ \sin k \end{pmatrix}$$

satisfies

$$f(x_{k+1}) < f(x_k)$$

for $k = 0, 1, 2, \dots$. Show that every point on the unit circle

$$\{x \mid |0x|_{0_2} = 1\}$$

is a limit point for $\{x_k\}$.

Hint: Every value $\theta \in [0, 2\pi]$ is a limit point of the subsequence $\{\xi_k\}$ defined by

$$\xi_k = k \pmod{2\pi} = k - 2\pi \left\lfloor \frac{k}{2\pi} \right\rfloor,$$

where the operator $\lfloor \cdot \rfloor$ denotes rounding down to the next integer.

SOLUTION. 由 $\cos^2 k + \sin^2 k = 1$, 有

$$f(x_k) = |0x_k|_{0_2} = 1 + 2^{-k}.$$

因此对任意 $k \geq 0$,

$$f(x_{k+1}) = 1 + 2^{-(k+1)} < 1 + 2^{-k} = f(x_k).$$

下面证明其极限点恰好覆盖单位圆。记

$$\xi_k = k \pmod{2\pi} \in [0, 2\pi).$$

由于 $1/(2\pi)$ 是无理数, 无理旋转的稠密性定理说明 $\{\xi_k : k \geq 0\}$ 在 $[0, 2\pi)$ 中稠密。为完整起见, 该结论可由抽屉原理得到: 对任意 N , 把 $0, 1, \dots, N$ 在模 2π 意义下的余数放入 N 个等长小区间, 必有两个余数之差的模长不超过 $2\pi/N$; 又因 2π 不是整数的有理倍数, 该模长不为零。让这一小旋转反复作用即可在圆周上形成间距不超过 $2\pi/N$ 的网, 从而逼近任意给定角度。令 $N \rightarrow \infty$ 即得稠密性。

任取单位圆上的点

$$\bar{x} = (\cos \theta, \sin \theta)^T, \quad \theta \in [0, 2\pi).$$

稠密性不仅对原序列成立，而且对任意尾序列 $\{\xi_k : k \geq K\}$ 也成立（尾序列只是原稠密轨道旋转了一个固定角度）。故可递归选取严格递增的指标 k_j ，使得

$$|\xi_{k_j} - \theta| < \frac{1}{j}.$$

于是 $\xi_{k_j} \rightarrow \theta$ ，同时 $1 + 2^{-k_j} \rightarrow 1$ 。利用正弦、余弦的连续性，

$$x_{k_j} = (1 + 2^{-k_j}) \begin{pmatrix} \cos \xi_{k_j} \\ \sin \xi_{k_j} \end{pmatrix} \rightarrow \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} = \bar{x}.$$

因此单位圆上的每一点都是 $\{x_k\}$ 的极限点。 □

PROBLEM III[2.6] Prove that all isolated local minimizers are strict.

Hint: Take an isolated local minimizer x^* and a neighborhood $\mathcal{N}$. Show that for any $x \in \mathcal{N}$, $x \neq x^*$, we must have

$$f(x) > f(x^*).$$

SOLUTION. 设 x^* 是孤立的局部极小点。由局部极小点的定义，存在开邻域 $\mathcal{N}_0$ ，使得

$$f(x) \geq f(x^*), \quad x \in \mathcal{N}_0.$$

又因为 x^* 是孤立的，所以可以缩小邻域，取 $\mathcal{N} \subseteq \mathcal{N}_0$ ，使得 $\mathcal{N}$ 中除 x^* 外不存在其他局部极小点。

若 x^* 不是严格局部极小点，则在任意这样的邻域中都存在 $\bar{x} \neq x^*$ 使 $f(\bar{x}) = f(x^*)$ 。取充分小的开邻域 $\mathcal{U}$ 使 $\bar{x} \in \mathcal{U} \subseteq \mathcal{N}_0$ 。对任意 $y \in \mathcal{U}$ ，

$$f(y) \geq f(x^*) = f(\bar{x}),$$

所以 $\bar{x}$ 本身也是局部极小点。这与 $\mathcal{N}$ 中不含其他局部极小点矛盾。因此存在 x^* 的一个邻域 $\mathcal{N}$ ，使得

$$f(x) > f(x^*), \quad x \in \mathcal{N}, \quad x \neq x^*,$$

即 x^* 是严格局部极小点。 □

PROBLEM IV[2.7] Suppose that

$$f(x) = x^T Q x,$$

where Q is an $n \times n$ symmetric positive semidefinite matrix. Show using the definition (1.4) that $f(x)$ is convex on the domain $\mathbb{R}^n$.

Hint: It may be convenient to prove the following equivalent inequality:

$$f(y + \alpha(x - y)) - \alpha f(x) - (1 - \alpha)f(y) \leq 0,$$

for all $\alpha \in [0, 1]$ and all $x, y \in \mathbb{R}^n$.

SOLUTION. 任取 $x, y \in \mathbb{R}^n$ 及 $\alpha \in [0, 1]$ 。直接展开得

$$\begin{aligned} & f(y + \alpha(x - y)) - \alpha f(x) - (1 - \alpha)f(y) \\ &= (\alpha x + (1 - \alpha)y)^T Q(\alpha x + (1 - \alpha)y) - \alpha x^T Qx - (1 - \alpha)y^T Qy \\ &= -\alpha(1 - \alpha)(x - y)^T Q(x - y). \end{aligned}$$

由于 $Q \succeq 0$, 有 $(x - y)^T Q(x - y) \geq 0$; 又因为 $\alpha(1 - \alpha) \geq 0$, 故上式不大于零。因此

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y),$$

这正是 f 在 $\mathbb{R}^n$ 上为凸函数的定义。 □

PROBLEM V[2.9] Consider the function

$$f(x_1, x_2) = (x_1 + x_2^2)^2.$$

At the point $x^T = (1, 0)$ we consider the search direction

$$p^T = (-1, 1).$$

Show that p is a descent direction and find all minimizers of the problem (2.10).

SOLUTION. 在 $x = (1, 0)^T$ 处,

$$\nabla f(x) = 2(x_1 + x_2^2) \begin{pmatrix} 1 \\ 2x_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}.$$

因而

$$\nabla f(x)^T p = (2, 0) \begin{pmatrix} -1 \\ 1 \end{pmatrix} = -2 < 0,$$

所以 p 是下降方向。

沿该方向的一维搜索函数为

$$\phi(\alpha) = f(x + \alpha p) = f(1 - \alpha, \alpha) = (1 - \alpha + \alpha^2)^2.$$

注意到

$$1 - \alpha + \alpha^2 = \left(\alpha - \frac{1}{2}\right)^2 + \frac{3}{4} > 0,$$

故

$$\phi'(\alpha) = 2(1 - \alpha + \alpha^2)(2\alpha - 1)$$

仅在 $\alpha = 1/2$ 处为零, 并且在该点左侧为负、右侧为正。因此无论问题 (2.10) 的可行域取 $\alpha > 0$ 还是 $\alpha \geq 0$, 其唯一极小化步长均为

$$\boxed{\alpha^* = \frac{1}{2}}, \quad \phi(\alpha^*) = \left(\frac{3}{4}\right)^2 = \frac{9}{16}.$$

相应的一维搜索极小点为 $x + \alpha^* p = (1/2, 1/2)^T$ 。 □

PROBLEM VI[2.13] (For this and the following three questions, refer to the material on “Rates of Convergence” in Section A.2 of the Appendix.)

Show that the sequence

$$x_k = \frac{1}{k}$$

is not Q-linearly convergent, though it does converge to zero. (This is called sublinear convergence.)

SOLUTION. 显然 $x_k = 1/k \rightarrow 0$ 。若该序列 Q-线性收敛, 则存在常数 $r \in (0, 1)$ 以及 K , 使得对所有 $k \geq K$,

$$\frac{|x_{k+1} - 0|}{|x_k - 0|} \leq r.$$

但事实上

$$\frac{|x_{k+1}|}{|x_k|} = \frac{k}{k+1} \rightarrow 1.$$

因此对任意 $r < 1$, 只要 k 足够大便有 $k/(k+1) > r$, 与 Q-线性收敛的定义矛盾。故该序列收敛到零, 但不是 Q-线性收敛; 这种收敛称为次线性收敛。□

PROBLEM VII[2.14] Show that the sequence

$$x_k = 1 + (0.5)^{2^k}$$

is Q-quadratically convergent to 1.

SOLUTION. 令误差 $e_k = |x_k - 1| = (0.5)^{2^k}$ 。则

$$e_{k+1} = (0.5)^{2^{k+1}} = \left((0.5)^{2^k}\right)^2 = e_k^2.$$

所以对所有 k 都有 $e_{k+1} \leq C e_k^2$, 其中可取 $C = 1$ 。因此 x_k Q-二次收敛到 1。□

PROBLEM VIII[2.15] Does the sequence

$$x_k = \frac{1}{k!}$$

converge Q-superlinearly? Q-quadratically?

SOLUTION. 此时极限为零, 误差就是 $e_k = x_k = 1/k!$ 。首先,

$$\frac{e_{k+1}}{e_k} = \frac{1}{k+1} \rightarrow 0,$$

故该序列 Q-超线性收敛。

另一方面, 若其 Q-二次收敛, 则应存在常数 $C > 0$, 使充分大的 k 满足 $e_{k+1} \leq C e_k^2$ 。然而

$$\frac{e_{k+1}}{e_k^2} = \frac{1/(k+1)!}{1/(k!)^2} = \frac{k!}{k+1} \rightarrow +\infty.$$

这个比值不可能被某个固定常数控制。因此该序列 Q-超线性收敛, 但不 Q-二次收敛。□

PROBLEM IX[2.16] Consider the sequence $\{x_k\}$ defined by

$$x_k = \begin{cases} \left(\frac{1}{4}\right)^{2^k}, & k \text{ even}, \\ x_{k-1}/k, & k \text{ odd}. \end{cases}$$

Is this sequence Q-superlinearly convergent? Q-quadratically convergent? R-quadratically convergent?

SOLUTION. 各项均为正且趋于零。分别考察奇偶两种转移。

当 k 为偶数时, $k+1$ 为奇数, 故

$$\frac{x_{k+1}}{x_k} = \frac{1}{k+1} \rightarrow 0.$$

当 k 为奇数时,

$$x_k = \frac{4^{-2^{k-1}}}{k}, \quad x_{k+1} = 4^{-2^{k+1}},$$

从而

$$\frac{x_{k+1}}{x_k} = k \cdot 4^{-(2^{k+1}-2^{k-1})} \rightarrow 0.$$

两条子序列上的误差比均趋于零, 所以 $\{x_k\}$ Q-超线性收敛到零。

它并不 Q-二次收敛。事实上, 当 k 为偶数时,

$$\frac{x_{k+1}}{x_k^2} = \frac{1}{(k+1)x_k} \rightarrow +\infty,$$

故不存在统一的二次收敛常数。

最后构造控制序列

$$v_k = 2^{-2^k}.$$

它满足 $v_{k+1} = v_k^2$, 因而 Q-二次收敛到零。若 k 为偶数, 则

$$x_k = 4^{-2^k} = 2^{-2^{k+1}} \leq 2^{-2^k} = v_k;$$

若 k 为奇数, 则

$$x_k = \frac{4^{-2^{k-1}}}{k} = \frac{2^{-2^k}}{k} \leq v_k.$$

故 $0 < x_k \leq v_k$ 对所有 $k \geq 1$ 成立。按照 R-收敛的定义, $\{x_k\}$ R-二次收敛。综上, 该序列 Q-超线性、非 Q-二次, 但 R-二次收敛。□

PROBLEM X[3.1] Program the steepest descent and Newton algorithms using the backtracking line search, Algorithm 3.1. Use them to minimize the Rosenbrock function (2.22). Set the initial step length $\alpha_0 = 1$ and print the step length used by each method at each iteration. First try the initial point

$$x_0 = (1.2, 1.2)^T$$

and then the more difficult starting point

$$x_0 = (-1.2, 1)^T.$$

SOLUTION. 采用 Armijo 回溯条件

$$f(x_k + \alpha_k p_k) \leq f(x_k) + c_1 \alpha_k \nabla f(x_k)^T p_k,$$

其中初始步长 $\alpha_0 = 1$, 并取 $c_1 = 10^{-4}$ 、回缩因子 $\rho = 1/2$ 。最速下降法取 $p_k = -\nabla f(x_k)$; Newton 法取 $p_k = -\nabla^2 f(x_k)^{-1} \nabla f(x_k)$ 。终止条件设为 $\|\nabla f(x_k)\|_2 \leq 10^{-8}$ 。完整程序如下; 程序中的输出语句会打印每次迭代实际接受的步长。

```

1 import numpy as np
2
3 def fun(x):
4     return 100.0 * (x[1] - x[0]**2)**2 + (1.0 - x[0])**2
5
6 def grad(x):
7     return np.array([
8         -400.0*x[0]*(x[1] - x[0]**2) + 2.0*(x[0] - 1.0),
9         200.0*(x[1] - x[0]**2)
10    ])
11
12 def hess(x):
13     return np.array([
14         [1200.0*x[0]**2 - 400.0*x[1] + 2.0, -400.0*x[0]],
15         [-400.0*x[0], 200.0]
16    ])
17
18 def backtracking(x, p, alpha0=1.0, rho=0.5, c1=1.0e-4):
19     alpha = alpha0
20     f0, slope0 = fun(x), grad(x) @ p
21     assert slope0 < 0.0
22     while fun(x + alpha*p) > f0 + c1*alpha*slope0:
23         alpha *= rho
24     return alpha
25
26 def solve(x0, method, tol=1.0e-8, maxit=200000):
27     x = np.asarray(x0, dtype=float).copy()
28     steps = []
29     for k in range(maxit):
30         g = grad(x)
31         if np.linalg.norm(g) <= tol:
32             return x, steps
33         if method == 0: # steepest descent
34             p = -g
35         elif method == 1: # Newton
36             p = np.linalg.solve(hess(x), -g)
37         else:
38             raise ValueError
39         alpha = backtracking(x, p)
40         print(method, k, alpha)
41         steps.append(alpha)
42         x = x + alpha*p
43     raise RuntimeError
44
45 for x0 in ([1.2, 1.2], [-1.2, 1.0]):
46     for method in (0, 1):
47         x, steps = solve(x0, method)
48         print(x0, method, len(steps), x, fun(x),
49               np.linalg.norm(grad(x)))

```

运行结果汇总如下（迭代次数等于接受步长的个数）：

x_0	方法	迭代次数	$f(x_{\text{final}})$	$ 0\nabla f(x_{\text{final}}) _2$
$(1.2, 1.2)^T$	最速下降	18691	6.1610×10^{-17}	9.9537×10^{-9}
$(1.2, 1.2)^T$	Newton	8	1.0883×10^{-25}	1.4360×10^{-11}
$(-1.2, 1)^T$	最速下降	19435	6.1066×10^{-17}	9.9968×10^{-9}
$(-1.2, 1)^T$	Newton	21	3.7440×10^{-21}	4.4733×10^{-10}

两次 Newton 实验所接受的完整步长序列分别为

$$(1.2, 1.2)^T : 1, \frac{1}{2}, 1, 1, 1, 1, 1, 1,$$

$$(-1.2, 1)^T : 1, \frac{1}{8}, 1, 1, 1, \frac{1}{4}, 1, 1, 1, \frac{1}{2}, 1, 1, 1, \frac{1}{2}, 1, 1, 1, 1, 1, 1.$$

最速下降法步数很大，故不在正文逐项抄录；程序会逐项打印。其步长主要为 $2^{-9} = 0.001953125$ ，这也解释了它在 Rosenbrock 狭长弯曲谷底中的缓慢收敛。□

PROBLEM XI[3.4] Show that the one-dimensional minimizer of a strongly convex quadratic function always satisfies the Goldstein conditions (3.11).

SOLUTION. 设一维强凸二次函数为

$$\phi(\alpha) = \phi(0) + b\alpha + \frac{1}{2}q\alpha^2, \quad q > 0.$$

在线搜索中 p 是下降方向，所以 $b = \phi'(0) < 0$ 。唯一极小化步长为

$$\alpha^* = -\frac{b}{q} > 0.$$

利用 $q\alpha^* = -b$ ，可得

$$\phi(\alpha^*) = \phi(0) + b\alpha^* + \frac{1}{2}q(\alpha^*)^2 = \phi(0) + \frac{1}{2}b\alpha^*.$$

Goldstein 条件为：对某个 $c \in (0, 1/2)$ ，

$$\phi(0) + (1-c)\alpha\phi'(0) \leq \phi(\alpha) \leq \phi(0) + c\alpha\phi'(0).$$

因为 $b\alpha^* < 0$ 且 $c < 1/2 < 1-c$ ，所以

$$(1-c)b\alpha^* \leq \frac{1}{2}b\alpha^* \leq cb\alpha^*.$$

两边同时加上 $\phi(0)$ ，恰好得到 α^* 满足上述两个 Goldstein 不等式。□

PROBLEM XII[3.5] Prove that

$$|0Bx|_0 \geq \frac{|0x|_0}{|0B^{-1}|_0}$$

for any nonsingular matrix B . Use this fact to establish (3.19).

SOLUTION. 由 B 非奇异及算子范数的次乘性,

$$|0x|0 = |0B^{-1}Bx|0 \leq |0B^{-1}|0 |0Bx|0.$$

整理即得

$$\boxed{|0Bx|0 \geq \frac{|0x|0}{|0B^{-1}|0}}.$$

下面用它证明 (3.19)。Newton 型方向满足 $B_k p_k = -\nabla f_k$, 其中 B_k 对称正定, 并假设

$$|0B_k|0 |0B_k^{-1}|0 \leq M.$$

将刚证明的不等式用于 $B_k^{1/2}$, 再平方, 得到

$$p_k^T B_k p_k = |0B_k^{1/2} p_k|0^2 \geq \frac{|0p_k|0^2}{|0B_k^{-1/2}|0^2} = \frac{|0p_k|0^2}{|0B_k^{-1}|0}.$$

另一方面, $|0B_k p_k|0 \leq |0B_k|0 |0p_k|0$ 。因此方向 p_k 与 $-\nabla f_k$ 之间夹角 θ_k 满足

$$\begin{aligned} \cos \theta_k &= \frac{-\nabla f_k^T p_k}{|0\nabla f_k|0 |0p_k|0} = \frac{p_k^T B_k p_k}{|0B_k p_k|0 |0p_k|0} \\ &\geq \frac{|0p_k|0^2 / |0B_k^{-1}|0}{|0B_k|0 |0p_k|0^2} = \frac{1}{|0B_k|0 |0B_k^{-1}|0} \geq \frac{1}{M}. \end{aligned}$$

这就是 (3.19)。 □

PROBLEM XIII[3.7] Prove the result (3.28) by working through the following steps.

First, use (3.26) to show that

$$|0x_k - x^*|0_Q^2 - |0x_{k+1} - x^*|0_Q^2 = 2\alpha_k \nabla f_k^T Q(x_k - x^*) - \alpha_k^2 \nabla f_k^T Q \nabla f_k,$$

where $|0 \cdot |0_Q$ is defined by (3.27).

Second, use the fact that

$$\nabla f_k = Q(x_k - x^*)$$

to obtain

$$|0x_k - x^*|0_Q^2 - |0x_{k+1} - x^*|0_Q^2 = 2 \frac{(\nabla f_k^T \nabla f_k)^2}{\nabla f_k^T Q \nabla f_k} - \frac{(\nabla f_k^T \nabla f_k)^2}{\nabla f_k^T Q \nabla f_k}$$

and

$$|0x_k - x^*|0_Q^2 = \nabla f_k^T Q^{-1} \nabla f_k.$$

SOLUTION. 对严格凸二次函数

$$f(x) = \frac{1}{2} x^T Q x - b^T x, \quad Q \succ 0,$$

记 $x^* = Q^{-1}b$, 则 $\nabla f_k = Q(x_k - x^*)$ 。最速下降法配合精确线搜索给出

$$x_{k+1} = x_k - \alpha_k \nabla f_k, \quad \alpha_k = \frac{\nabla f_k^T \nabla f_k}{\nabla f_k^T Q \nabla f_k}.$$

令 $e_k = x_k - x^*$ 。按照 $|0z|0_Q^2 = z^T Q z$ 的定义,

$$\begin{aligned} |0e_k|0_Q^2 - |0e_{k+1}|0_Q^2 &= e_k^T Q e_k - (e_k - \alpha_k \nabla f_k)^T Q (e_k - \alpha_k \nabla f_k) \\ &= 2\alpha_k \nabla f_k^T Q e_k - \alpha_k^2 \nabla f_k^T Q \nabla f_k. \end{aligned}$$

又因为 $Qe_k = \nabla f_k$, 所以

$$\nabla f_k^T Q e_k = \nabla f_k^T \nabla f_k.$$

代入精确线搜索步长, 得到

$$\begin{aligned} |0e_k|0_Q^2 - |0e_{k+1}|0_Q^2 &= 2 \frac{(\nabla f_k^T \nabla f_k)^2}{\nabla f_k^T Q \nabla f_k} - \frac{(\nabla f_k^T \nabla f_k)^2}{\nabla f_k^T Q \nabla f_k} \\ &= \frac{(\nabla f_k^T \nabla f_k)^2}{\nabla f_k^T Q \nabla f_k}. \end{aligned}$$

同时 $e_k = Q^{-1} \nabla f_k$, 故

$$|0e_k|0_Q^2 = e_k^T Q e_k = \nabla f_k^T Q^{-1} \nabla f_k.$$

将差值等式除以 $|0e_k|0_Q^2$, 可得所要求的 (3.28):

$$|0x_{k+1} - x^*|0_Q^2 = \left[1 - \frac{(\nabla f_k^T \nabla f_k)^2}{(\nabla f_k^T Q \nabla f_k)(\nabla f_k^T Q^{-1} \nabla f_k)} \right] |0x_k - x^*|0_Q^2.$$

再结合题 3.8 的 Kantorovich 不等式, 还可推出

$$|0x_{k+1} - x^*|0_Q \leq \frac{\lambda_n - \lambda_1}{\lambda_n + \lambda_1} |0x_k - x^*|0_Q.$$

□

PROBLEM XIV[3.8] Let Q be a positive definite symmetric matrix. Prove that for any vector x , we have

$$\frac{(x^T x)^2}{(x^T Q x)(x^T Q^{-1} x)} \geq \frac{4\lambda_n \lambda_1}{(\lambda_n + \lambda_1)^2},$$

where λ_n and λ_1 are, respectively, the largest and smallest eigenvalues of Q .

(This relation, which is known as the Kantorovich inequality, can be used to deduce (3.29) from (3.28).)

SOLUTION. 当 $x = 0$ 时不等式显然成立。以下设 $x \neq 0$ 。对 Q 作正交谱分解 $Q = U \Lambda U^T$, 其中

$$0 < \lambda_1 \leq \lambda_i \leq \lambda_n.$$

令 $z = U^T x$ 及

$$w_i = \frac{z_i^2}{\sum_j z_j^2},$$

则 $w_i \geq 0$ 且 $\sum_i w_i = 1$ 。记

$$A = \sum_i w_i \lambda_i, \quad B = \sum_i \frac{w_i}{\lambda_i}.$$

由 $\lambda_i \in [\lambda_1, \lambda_n]$, 有

$$(\lambda_i - \lambda_1)(\lambda_n - \lambda_i) \geq 0,$$

等价地,

$$\lambda_i + \frac{\lambda_1 \lambda_n}{\lambda_i} \leq \lambda_1 + \lambda_n.$$

以 w_i 加权求和, 得到

$$A + \lambda_1 \lambda_n B \leq \lambda_1 + \lambda_n.$$

再由算术-几何平均不等式,

$$4\lambda_1 \lambda_n AB \leq (A + \lambda_1 \lambda_n B)^2 \leq (\lambda_1 + \lambda_n)^2.$$

另一方面, 正交变换保持内积, 故

$$A = \frac{x^T Q x}{x^T x}, \quad B = \frac{x^T Q^{-1} x}{x^T x}.$$

于是

$$\frac{(x^T Q x)(x^T Q^{-1} x)}{(x^T x)^2} = AB \leq \frac{(\lambda_n + \lambda_1)^2}{4\lambda_n \lambda_1}.$$

所有量均为正, 取倒数便得

$$\boxed{\frac{(x^T x)^2}{(x^T Q x)(x^T Q^{-1} x)} \geq \frac{4\lambda_n \lambda_1}{(\lambda_n + \lambda_1)^2}.$$

□

`ROBLEM XV[3.9]` Program the BFGS algorithm using the line search algorithm described in this chapter that implements the strong Wolfe conditions. Have the code verify that

$$y_k^T s_k$$

is always positive. Use it to minimize the Rosenbrock function using the starting points given in Exercise 3.1.

SOLUTION. 强 Wolfe 条件为

$$f(x_k + \alpha_k p_k) \leq f(x_k) + c_1 \alpha_k \nabla f_k^T p_k,$$

$$|\nabla f(x_k + \alpha_k p_k)^T p_k| \leq c_2 |\nabla f_k^T p_k|,$$

这里取 $c_1 = 10^{-4}$ 、 $c_2 = 0.9$ 。程序使用试探区间扩张和二分 zoom 实现强 Wolfe 线搜索, 并直接更新逆 Hessian 近似 H_k :

$$H_{k+1} = (I - \rho_k s_k y_k^T) H_k (I - \rho_k y_k s_k^T) + \rho_k s_k s_k^T, \quad \rho_k = (y_k^T s_k)^{-1}.$$

以下代码接续题 3.1 中的 `fun` 和 `grad` 定义。

```

1 def strongWolfe(x, p, c1=1.0e-4, c2=0.9):
2     phi0 = fun(x)
3     dphi0 = grad(x) @ p
4     assert dphi0 < 0.0
5
6     def phi(alpha):
7         return fun(x + alpha*p)
8
9     def dphi(alpha):
10        return grad(x + alpha*p) @ p
11
12    def zoom(alphaLo, alphaHi):
13        phiLo = phi(alphaLo)
14        for zoomIter in range(100):
15            alpha = 0.5*(alphaLo + alphaHi)
16            phiAlpha = phi(alpha)
17            if (phiAlpha > phi0 + c1*alpha*dphi0 or
18                phiAlpha >= phiLo):
19                alphaHi = alpha
20            else:
21                dphiAlpha = dphi(alpha)
22                if abs(dphiAlpha) <= -c2*dphi0:
23                    return alpha
24                if dphiAlpha*(alphaHi - alphaLo) >= 0.0:
25                    alphaHi = alphaLo
26                alphaLo = alpha
27                phiLo = phiAlpha
28        raise RuntimeError
29
30    alphaPrev, phiPrev = 0.0, phi0
31    alpha = 1.0
32    for i in range(100):
33        phiAlpha = phi(alpha)
34        if (phiAlpha > phi0 + c1*alpha*dphi0 or
35            (i > 0 and phiAlpha >= phiPrev)):
36            return zoom(alphaPrev, alpha)
37        dphiAlpha = dphi(alpha)
38        if abs(dphiAlpha) <= -c2*dphi0:
39            return alpha
40        if dphiAlpha >= 0.0:
41            return zoom(alpha, alphaPrev)
42        alphaPrev, phiPrev = alpha, phiAlpha
43        alpha *= 2.0
44    raise RuntimeError
45
46 def bfgs(x0, tol=1.0e-8, maxit=10000):
47     x = np.asarray(x0, dtype=float).copy()
48     H = np.eye(len(x))
49     history = []
50     for k in range(maxit):
51         g = grad(x)
52         if np.linalg.norm(g) <= tol:
53             return x, history
54         p = -H @ g
55         alpha = strongWolfe(x, p)
56         s = alpha*p
57         xNew = x + s
58         y = grad(xNew) - g
59         ys = y @ s
60         assert ys > 0.0
61         history.append((alpha, ys))
62         print(k, alpha, ys)
63
64         rho = 1.0/ys
65         I = np.eye(len(x))
66         H = ((I - rho*np.outer(s, y)) @ H @

```

```

67     (I - rho*np.outer(y, s)) + rho*np.outer(s, s))
68     x = xNew
69     raise RuntimeError
70
71 for x0 in ([1.2, 1.2], [-1.2, 1.0]):
72     x, history = bfgs(x0)
73     print(x0, len(history), x, fun(x), np.linalg.norm(grad(x)),
74           min(ys for alpha, ys in history))

```

实验结果为

x_0	迭代次数	$f(x_{\text{final}})$	$ 0\nabla f(x_{\text{final}}) _2$	$\min_k y_k^T s_k$
$(1.2, 1.2)^T$	17	2.1728×10^{-20}	3.0773×10^{-9}	6.0567×10^{-17}
$(-1.2, 1)^T$	35	1.2287×10^{-22}	4.7420×10^{-10}	3.7294×10^{-18}

两组实验中所有 $y_k^T s_k$ 均严格为正。事实上，这不仅是数值现象：由 $s_k = \alpha_k p_k$ 和强 Wolfe 曲率条件可得

$$\begin{aligned}
 y_k^T s_k &= \alpha_k (\nabla f_{k+1}^T p_k - \nabla f_k^T p_k) \\
 &\geq \alpha_k (c_2 - 1) \nabla f_k^T p_k > 0,
 \end{aligned}$$

其中最后一个不等式使用了 $\nabla f_k^T p_k < 0$ 及 $c_2 < 1$ 。因此只要 $H_0 \succ 0$ ，BFGS 更新便会保持 $H_k \succ 0$ 。□

PROBLEM XVI[3.12] Consider a block diagonal matrix B with 1×1 and 2×2 blocks. Show that the eigenvalues and eigenvectors of B can be obtained by computing the spectral decomposition of each diagonal block separately.

SOLUTION. 把各对角块记为 $B_1, \dots, B_m$ 。其中 1×1 块本身就是一个特征值；对每个实对称的 2×2 块分别作谱分解

$$B_j = U_j \Lambda_j U_j^T, \quad U_j^T U_j = I.$$

令

$$U = \text{diag}(U_1, \dots, U_m), \quad \Lambda = \text{diag}(\Lambda_1, \dots, \Lambda_m),$$

便有

$$U^T U = I, \quad B = U \Lambda U^T.$$

因此各小块的特征值合在一起（按代数重数计数）就是 B 的全部特征值。若 u 是第 j 个块的特征向量，把 u 放在第 j 个坐标块、其余坐标补零，所得向量 $\tilde{u}$ 满足

$$B \tilde{u} = \lambda \tilde{u},$$

因而就是整个矩阵的特征向量。即使同一特征值出现在不同块中，来自这些块的补零向量也彼此正交，并共同张成相应特征子空间。

具体地，对称块

$$C = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$$

的两个特征值可单独计算为

$$\lambda_{\pm} = \frac{a + d \pm \sqrt{(a - d)^2 + 4b^2}}{2},$$

再解 $(C - \lambda_{\pm}I)u_{\pm} = 0$ 并单位化即可得到该块的正交特征向量。所以计算 B 的谱分解完全可以分解为若干个互不耦合的 1×1 或 2×2 小问题。 $\square$